Safety as a capability learned through shared reasoning

Safety requests

General tasks

Shared reasoning structure

Intent / goal
Understand the request

Knowledge
Apply relevant facts and constraints

Consequences
Reason about possible outcomes

Stronger safety capability with general capability preserved

Safety retained during general-only SFT and GRPO without safety data or rewards

Jointly learned during safety mid-training

Constructing the safety mid-training corpus

Prompt
Safety label
Reference answer

Structured document

Task
Core goal
Relevant knowledge
Key constraints
Possible consequences
Best feasible path
Verification

Base model generates

Reasoning trace and answer

Grounded in the document and reference answer

Judge filtering

Faithfulness
Coherence
Answer consistency

Pass rubric checks
Quality score >= 4

Training records

Document prose
+
Prompt, reasoning and answer

Safety mid-training

Continue training the base model

Self-generated documents and traces; both record types are filtered before training.